

4. Troubleshooting

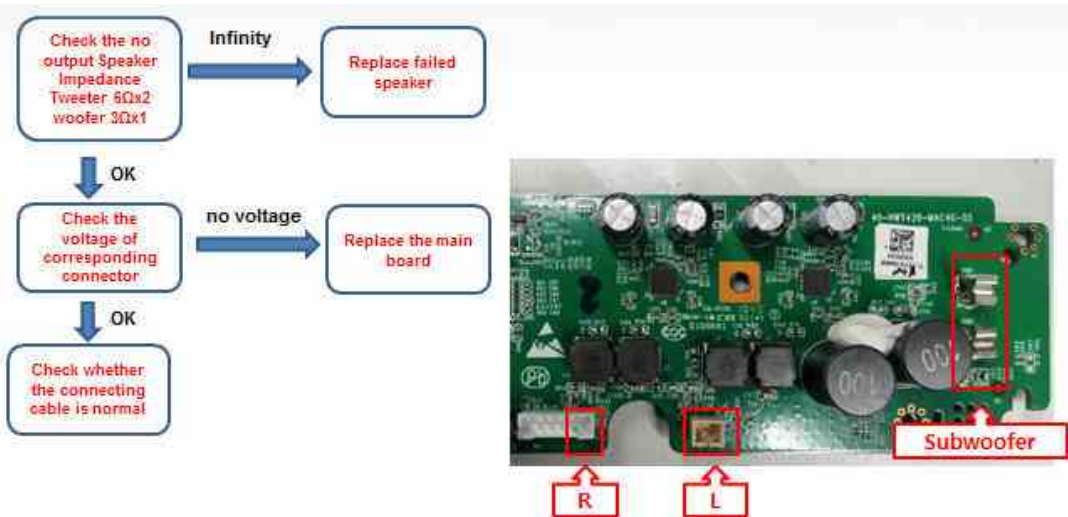
4-1. Checkpoints by Error Mode

Oscilloscope Setting Values	Check DC voltage	Check I ² C& I ² S signal	check PWM signal	Check L+,L- signal
Voltage/DIV	Depend voltage	1V/div	10V/div	10V/div
TIME/DIV	500ms/div	20us/div	1us/div	500us/div

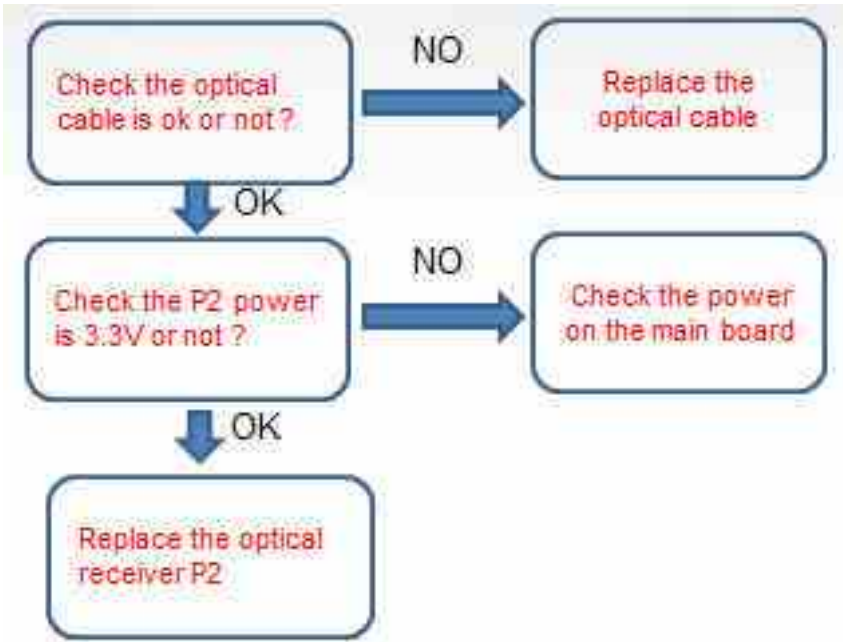
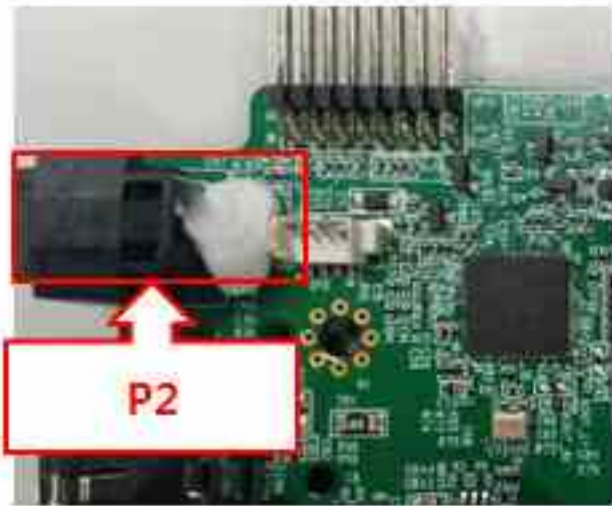
4-1-1. No Power

Symptom	- If you connect the power cables, the LED on the front will turn on - If the units do not work
Major checkpoints	In case of no power
Diagnostics	<pre> graph TD A[Check pin 22 to 24 voltage of XP1 on the main board] -- "< 22V" --> B[Check pin 1/2 to 3/4 voltage of XP6 on the power board] A -- "22-24V" --> C[Check the output of CVB4 on the main board] B -- "no voltage" --> D[Check Is the power cord connected properly] B -- "AC voltage" --> E[Replace the power board] C -- "< 4.5V" --> F[Replace the UB1 on main board] C -- "4.5-5.5V" --> G[Check the output of CBK1 on the main board] G -- "< 3.1V" --> H[Replace the UBK1 on main board] G -- "3.1-3.5V" --> I[U6 on the main board fault Replace the main board] I -- "NO" --> J[Replace the VFD Board] </pre> The flowchart provides a systematic approach to diagnosing power issues. It begins by checking the input voltage at XP1. If it's low, the power board (XP6) is the likely culprit. If the input is correct, the fault lies on the main board. The next steps involve checking the CVB4 output, then the CBK1 output, and finally the U6 component. Each step has a specific voltage range for a 'pass' and a replacement action for a 'fail'. The final step, if all other components are checked and the issue persists, is to replace the VFD Board. Two photographs are included to identify components: the top photo shows the power board connector XP6, and the bottom photo shows the main board components U6, CBK1, CVB4, UB1, UBK1, and XP1.
Caution	Before working, disconnect the power cord.

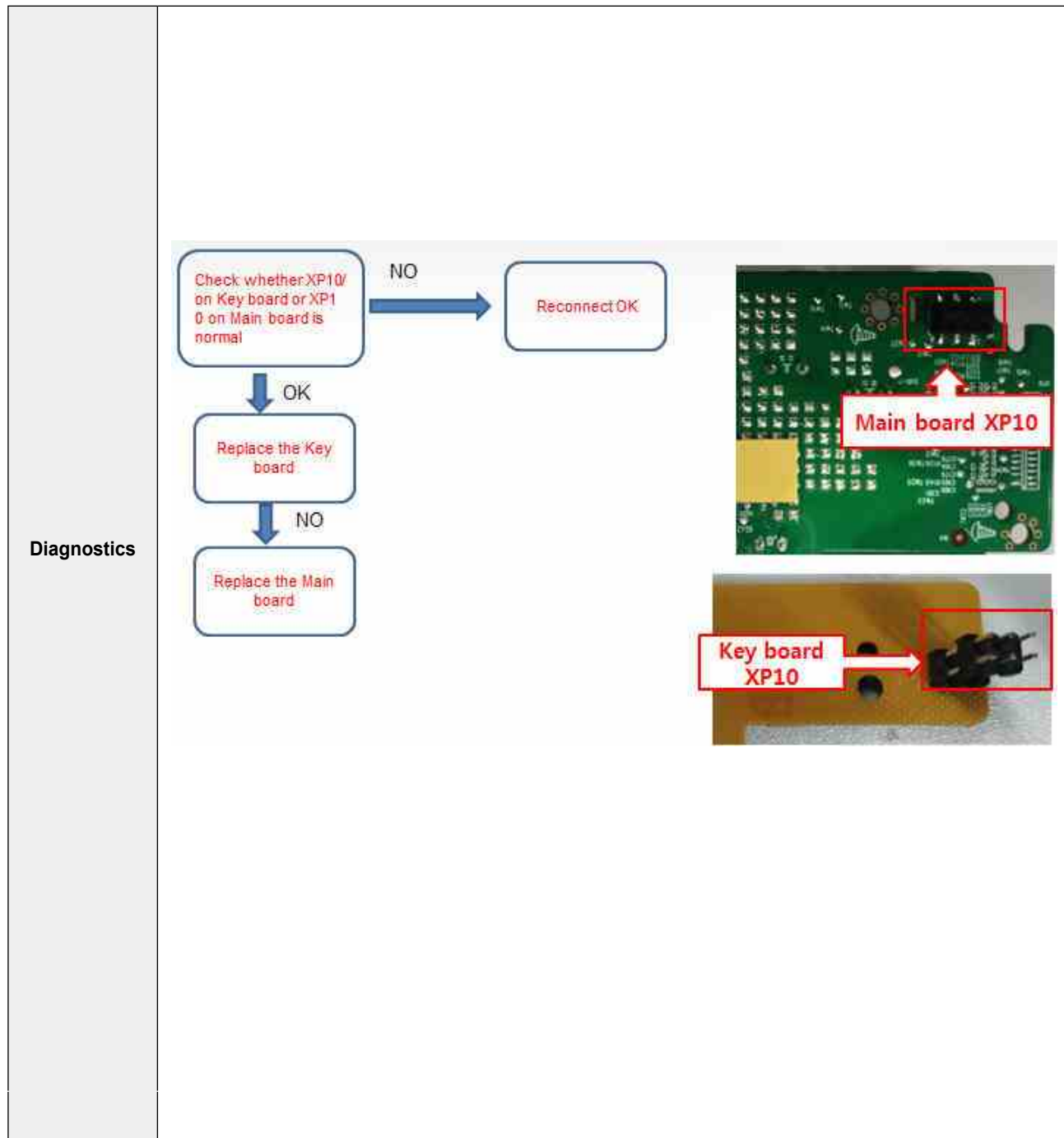
4-1-2. No Audio Output

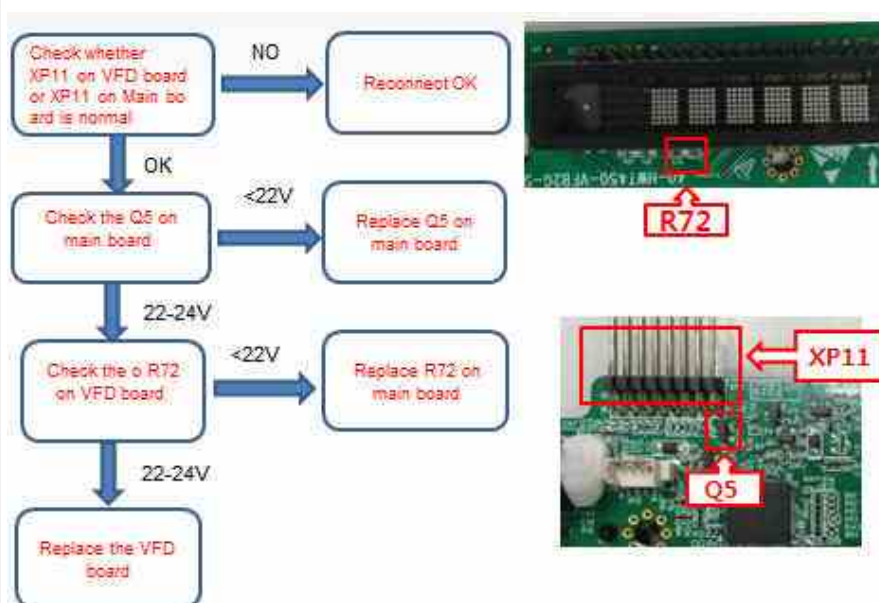
Symptom	In case of no sound
Major checkpoints	- Speaker connector is not connected properly or check for damage. - Make sure your sound processing functions operate properly on the motherboard. - Make sure the speaker isn't bad.
Diagnostics	 The diagnostics section contains a flowchart and a photograph of a motherboard. The flowchart starts with a box 'Check the no output Speaker Impedance Tweeter 6Ωx2 woofer 3Ωx1'. An arrow labeled 'Infinity' points to 'Replace failed speaker'. An arrow labeled 'OK' points down to 'Check the voltage of corresponding connector'. An arrow labeled 'no voltage' points to 'Replace the main board'. An arrow labeled 'OK' points down to 'Check whether the connecting cable is normal'. To the right is a photograph of a green motherboard. Red boxes highlight the 'R' and 'L' speaker connectors at the bottom, and a 'Subwoofer' connector on the right side.
Caution	Before working, disconnect the power cord.

4-1-3. Optical is not working

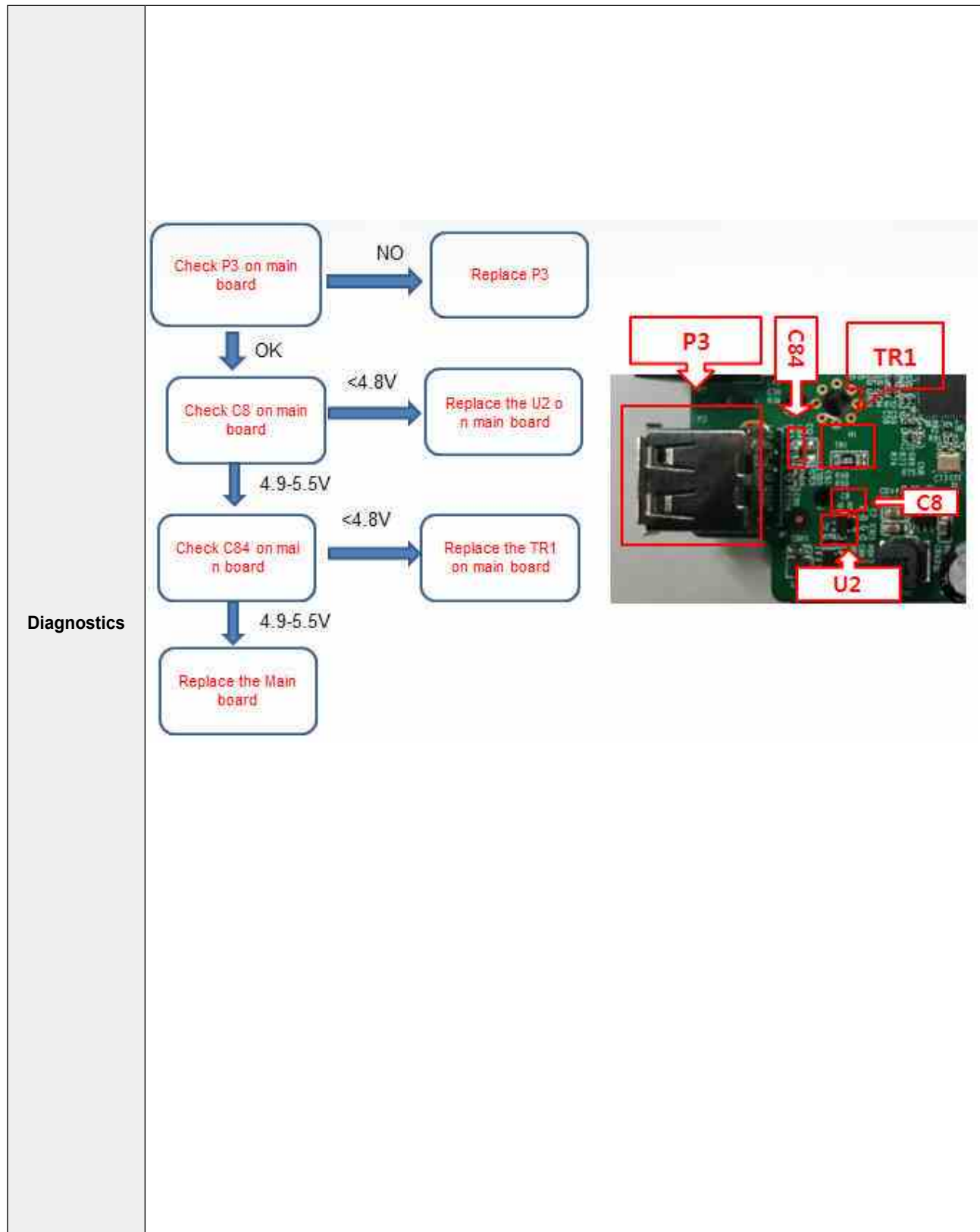
Symptom	In case of optical is not working
Major checkpoints	
Diagnostics	 <pre> graph TD A[Check the optical cable is ok or not?] -- NO --> B[Replace the optical cable] A -- OK --> C[Check the P2 power is 3.3V or not?] C -- NO --> D[Check the power on the main board] C -- OK --> E[Replace the optical receiver P2] </pre> 
Caution	Before working, disconnect the power cord.

4-1-4. In case of key is not working



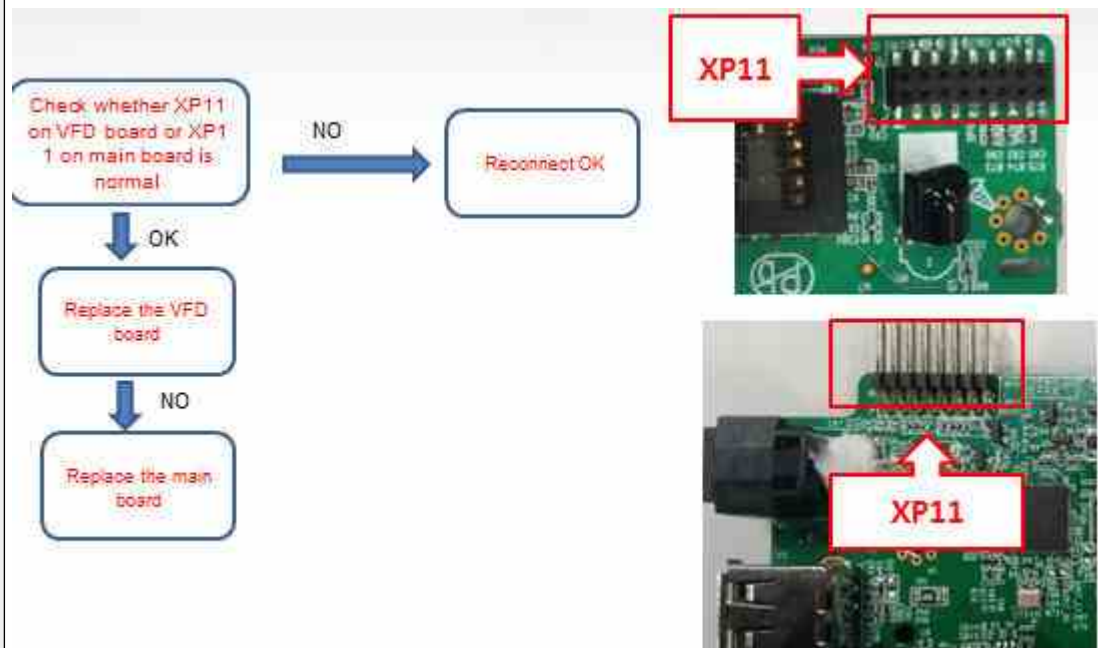
4-1-5. In case of VFD is not working**Diagnostics**

4-1-6. In case of USB is not working



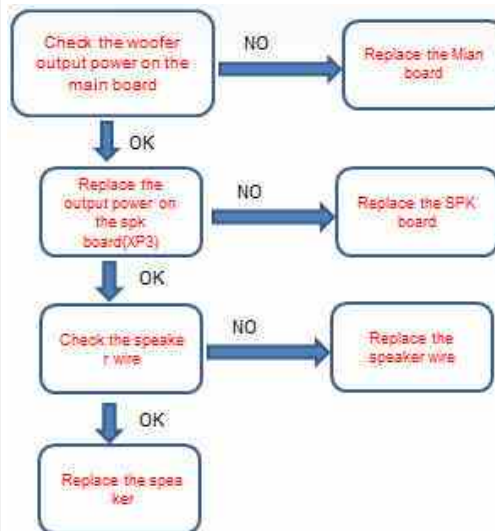
4-1-7. In case Bluetooth is not working

Diagnostics



4-1-8. In case woofer have no power output

Diagnostics



4-2. Measures to be taken when the Protection Circuit operates

4-2-1. AMP Pre-Inspection relating to Power Protection

If you think there are problems at the AMP PCB, you can check the PCB without disassembling the set by following the test below.

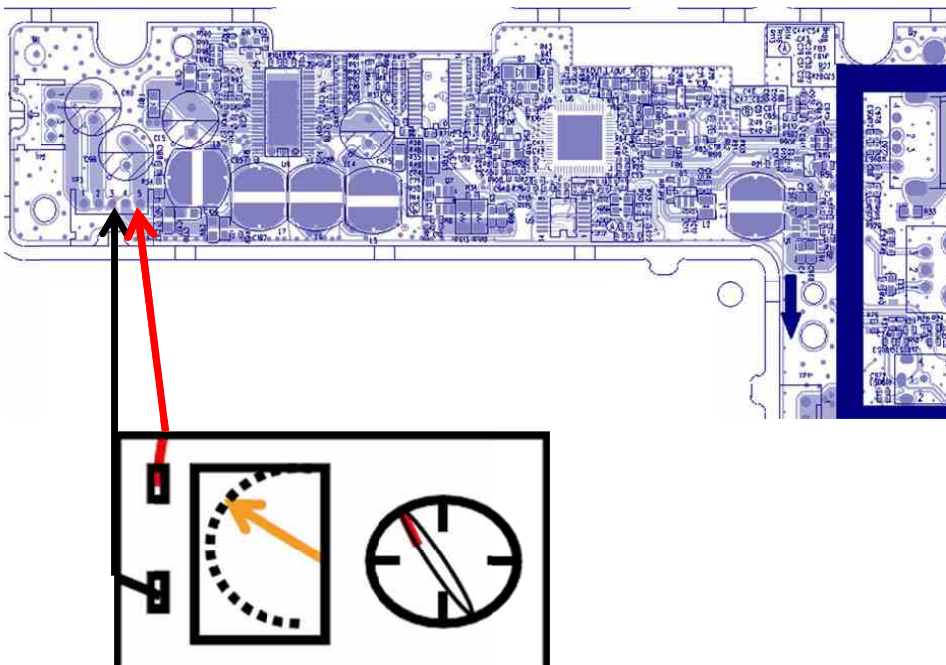


Caution

- Do not connect the power cord during the test!

Resistance using Tester	
F/R CH	9.48M ohm

- If there is a large difference than the value listed above then the AMP PCB has a problem.



4-3. Initialization & Update

4-3-1. Set Initialization method

Initialization

Service or Factory Initial: clear BT link and use history

1. Turn the unit on
2. Press and hold the “+” and “-” button on Sound bar more than 5s
3. “INIT” appear on VFD , about 3s.
4. If initialization successful, “OK” appear on VFD and then unit turn off.

Customer Initial: no clear BT link and use history

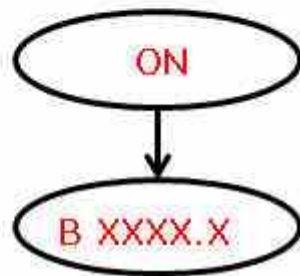
1. Turn the unit off
2. Press and hold the “PLAY/PAUSE” button on Remote or Sound bar more than 5s
3. If initialization successful, “INIT” appear on VFD , about 3s.

4-3. Initialization & Update

4-3-2. How to check firmware version

. Software check method

- Sound Bar(using the remote)
- 1) Prepare the remote control.
- 2) Turn the unit off
- 3) Press the "  " button on the remote for 5 seconds.
- 4) OLED will display each versions like below sequence.



B : Backend version(Representative Ver.)

4-3-3. USB Update procedure

COPY Update File:

- ◆ Copy update file to USB root, Make sure no others file in USB root.
- ◆ File need to be named as below table shows.

Module	Updated SW file name
HW-T420	HMS-T4X0WWB-xxxx.ROM

Steps:

1. Clear USB flash driver, Copy "HMS-T4X0WWB-xxxx.ROM" into USB flash driver root,
2. Plug USB flash driver into USB terminal of bar, power on T450 sound bar
3. Switch source to USB, then VFD will display "UPDATE" and an progress form 0~100.
4. After SW update is finished, Bar will restart automatically
5. [Check version](#), (click and jump to chapter check version) the bar version shall be "B xxxx"

ashakaas